TOPIC 5: MICROCOMPUTER APPLICATIONS

Official Undergraduate Lecture Slide Deck (Academic Year 2026/2027)

- Curriculum Coverage: Word Processors (5.1) | Spreadsheets (5.2) | Web & Email Protocols (5.3)
- **Course Instructor: Madam Bertha Msuliche Lebalwa (MSc)**
- Assessment Structure: 2 Assignments (5%), 2 Tests (15%), Practical Report (10%), Practical Test (30%), UE (40%)*

**Assessment weightings are subject to adjustment by the course instructor.*

Topic 5: Lecture Outline & Learning Roadmap

Productivity Applications, Spreadsheets & Web Systems (Coverage: 5.1, 5.2, 5.3)

- Section 5.1: Word Processing Systems & Academic Document Engineering
  - Structural document architecture: Character/paragraph formatting, section breaks, margins
  - Advanced automation: Heading styles, automated Table of Contents (TOC), cross-references
  - Mail merge architecture: Data sources, merge fields, and conditional rule filters (`IF...THEN`)
- Section 5.2: Spreadsheet Applications & Simple Calculations
  - Worksheet layout: rows, columns, cells, and cell references
  - Simple arithmetic formulas and introductory `SUM` and `AVERAGE` functions
  - Organizing information with clear labels, basic sorting, and charts
- Section 5.3: Web Browsing, Internet Protocols & Electronic Mail Architecture
  - World Wide Web: URLs, HTTP/HTTPS request-response cycles, status codes, and TLS certificates
  - Email protocols: SMTP (message submission/relay), POP3 vs IMAP4 (retrieval and sync)
  - Email security & anti-spoofing standards: SPF, DKIM, and DMARC

APPLICATION FOCUS

Use common productivity applications and communicate information clearly in academic and workplace settings.

Topic 5: Intended Learning Outcomes (ILOs)

Core Productivity & Communications Competencies

- Engineer complex multi-section academic documents with dynamic numbering and automated tables of contents.
- Design automated mail merge pipelines for dynamic reporting and student notifications.
- Identify rows, columns, cells, and cell references in a worksheet.
- Enter simple addition, subtraction, multiplication, and division formulas and use `SUM` or `AVERAGE`.
- Organize small worksheets using clear headings, basic sorting, and simple charts.
- Dissect Uniform Resource Locators (URLs) and trace HTTP/HTTPS client-server request-response cycles.
- Contrast POP3 and IMAP4 mail retrieval protocols and formulate SPF/DKIM anti-spoofing policies.

Word Processing: Academic Document Engineering

Beyond Basic Typing: Structural Typography and Sectioning

• Document Architecture: Hierarchical layers (Document → Sections → Paragraphs → Characters).

• Section Breaks vs Page Breaks:

- Page Break: Simply forces following text onto a new physical page; retains identical margins, orientation, and headers/footers.

- Section Break (Next Page): Divides document into independent formatting zones with distinct margins, page numbering formats, and orientations (Portrait vs Landscape).

• Thesis Compilation Rule at SUA:

- Section 1 (Preliminary Pages): Roman numerals (i, ii, iii) centered at footer.

- Section 2 (Chapter 1 onwards): Arabic numerals (1, 2, 3) aligned at top-right.

- Unlink Headers/Footers: 'Link to Previous' MUST be disabled between sections.

Automated Academic Referencing & Tables of Contents

Leveraging Semantic Heading Styles for Document Integrity

• The Semantic Heading Standard: Never manually format headings by changing font size and bolding.

• Heading Styles (Heading 1, Heading 2, Heading 3):

- Provides semantic structure recognized by document outline compilers.
- Enables one-click generation of the automated Table of Contents (TOC).

• Dynamic Updates: Modifying a chapter heading automatically updates the TOC and page numbers across a 200-page document via `F9` (Update Field).

• Table of Figures & Captions: Figures and tables tagged with official captions (`Insert Caption`), generating automated lists of figures and tables.

• Cross-Referencing: Dynamic links to numbered headings or figures that automatically track changes in figure numbering.

Mail Merge Architecture & Enterprise Dispatches

Mass Customization of Letters, Transcripts, and Advisory Notices

• Core Architecture: Combines a standardized template with a structured transactional data source.

• The Three Components of Mail Merge:

1. Main Document: Standardized text containing boilerplate prose and merge field placeholders (e.g., `«First_Name»`, `«Balance_Due»`).

2. Data Source: Structured tabular dataset (Excel spreadsheet, Access database, CSV file) where columns represent fields and rows represent records.

3. Merged Output Document: Consolidated multi-page file or direct email dispatch containing individual customized records.

• Conditional Logic Rules: Using Word `IF...THEN...ELSE` fields to dynamically insert customized advice based on numerical thresholds (e.g., 'Warning: Overdue Payment' if `Balance > 0`).

Spreadsheet Rows, Columns, and Cells

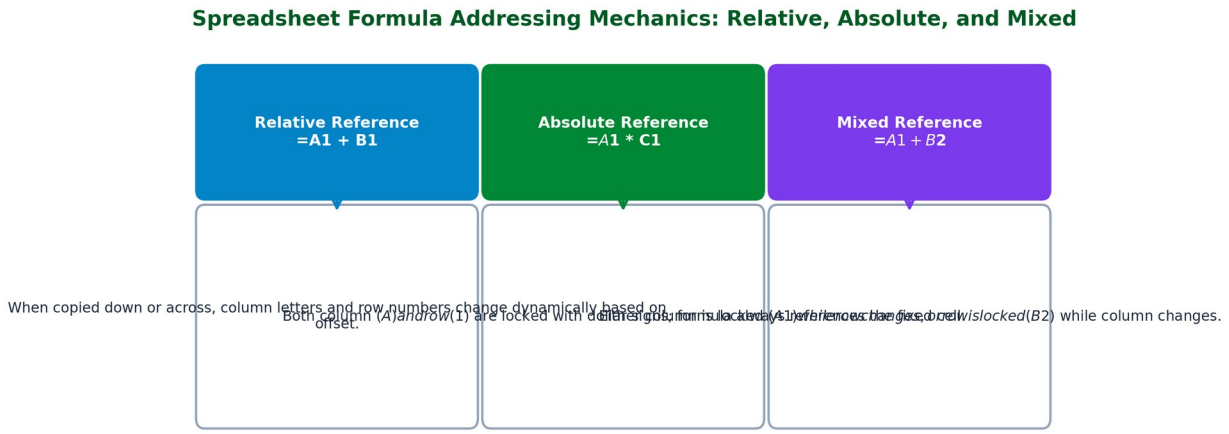


Figure: A worksheet is organized as columns, rows, and individual cells.

ARCHITECTURAL ANALYSIS & PRINCIPLES

- Columns are identified by letters and rows by numbers.
- A cell reference combines its column and row, such as `B2`.
- A formula can refer to values in other cells rather than repeating the values.
- When a simple formula is copied to another row, its cell references normally adjust to that row.
- Use clear headings so users can identify what each cell contains.

Simple Spreadsheet Formulas

A first calculation using farm data

- A formula begins with an equals sign and uses cell references.
- Addition: `=B2+C2`; subtraction: `=B2-C2`.
- Multiplication: `=B2*C2`; division: `=B2/C2`.
- Example: if B2 contains harvested kilograms and C2 contains the number of containers, `=B2/C2` gives the average per container.
- Press Enter to display the result. Check that the selected cells contain the intended values.

Using SUM and AVERAGE

Built-in functions for introductory spreadsheet tasks

- `=SUM(B2:B6)` adds the values in the selected range.
- `=AVERAGE(B2:B6)` finds the mean of the values in the selected range.
- Select the data range carefully and keep a clear label beside each result.
- A spreadsheet updates a result when a value used by its formula changes.
- Use these functions with small class examples such as crop yields, seedling counts, or animal feed records.

Checking and Presenting Spreadsheet Results

Simple review habits for class data

- Confirm that headings identify the information in each column.
- Use Sort to arrange records by a field such as date, crop, or animal ID.
- Use a simple chart to show a small set of values clearly.
- Check formulas and source data before sharing or printing a worksheet.
- Do not use a chart to imply a conclusion that the data does not support.

Sorting and Charting Small Data Sets

Basic ways to organize and present worksheet information

- Arrange related values in a labeled table, with one record on each row.
- Use the Sort command to arrange records by a chosen column.
- Use a basic chart when it helps communicate a small set of worksheet values.
- Keep a copy of the original data before reorganizing a worksheet.
- Explain what the chart shows and identify any limitations in the data.

The World Wide Web: Architecture & URLs

The Global Distributed Hypermedia System

• Core Architecture: Client-server hypermedia infrastructure running over TCP/IP, proposed by Tim Berners-Lee (1989).

• Anatomy of a Uniform Resource Locator (URL):

`https://www.sua.ac.tz:443/academics/courses?dept=diit&level=100#syllabus`

- Protocol / Scheme: `https://` (HyperText Transfer Protocol Secure).
- Fully Qualified Domain Name (FQDN): `www.sua.ac.tz` (resolved via DNS to server IP).
- Port Number: `:443` (standard encrypted TLS port; HTTP defaults to 80).
- Path: `/academics/courses` (hierarchical resource location on web server).
- Query String: `?dept=diit&level=100` (key-value parameters passed to backend script).
- Fragment / Anchor: `#syllabus` (client-side jump target within web page).

The HTTP/HTTPS Request-Response Cycle

Stateless Communication and RESTful Operations

- Client Request: Web browser opens TCP connection (port 80/443), sends HTTP request header:
  - HTTP Methods: `GET` (retrieve resource), `POST` (submit data/payload), `PUT` (replace resource), `DELETE` (remove resource).
- Server Response: Server processes request and returns HTTP status code, response headers, and payload (HTML/JSON).
- Standard HTTP Status Code Classes:
  - 1xx (Informational): Request received, continuing process.
  - 2xx (Success): `200 OK`, `201 Created`.
  - 3xx (Redirection): `301 Moved Permanently`, `302 Found`.
  - 4xx (Client Error): `400 Bad Request`, `401 Unauthorized`, `403 Forbidden`, `404 Not Found`.
  - 5xx (Server Error): `500 Internal Server Error`, `502 Bad Gateway`, `503 Service Unavailable`.

Electronic Mail Protocol Architecture

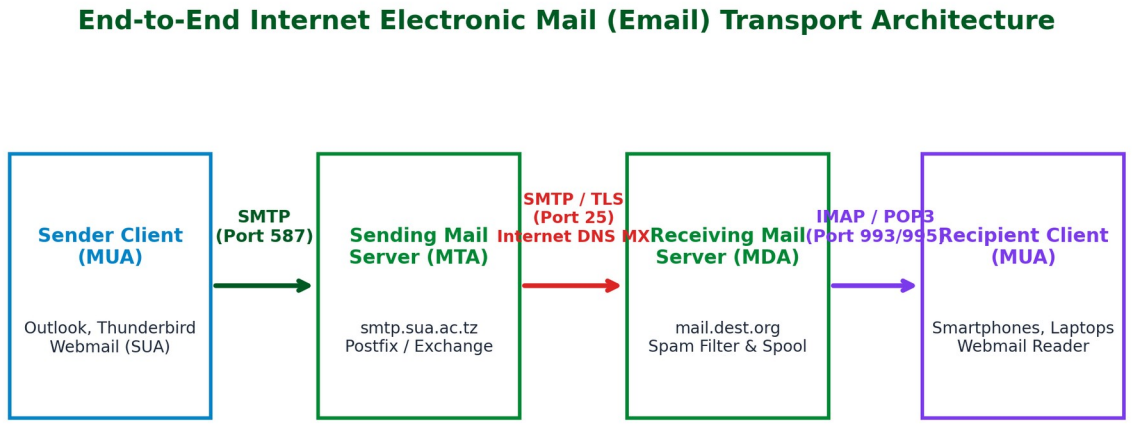


Figure: The complete email pipeline: MUA, MSA, MTA, MDA, SMTP, POP3, and IMAP4.

ARCHITECTURAL ANALYSIS & PRINCIPLES

- Sender MUA (Outlook/Thunderbird) submits email to Mail Submission Agent (MSA) via SMTP (Port 587 with STARTTLS).
- Mail Transfer Agent (MTA - Postfix/Sendmail) performs DNS MX record lookup and relays message across internet via SMTP (Port 25).
- Receiving MTA transfers message to Mail Delivery Agent (MDA - Dovecot), storing it in the user's mailbox.
- Recipient MUA retrieves or synchronizes mail via POP3 (Port 995) or IMAP4 (Port 993).

Email Retrieval: POP3 vs IMAP4

Store-and-Forward Download vs Real-Time Server Synchronization

- Post Office Protocol version 3 (POP3 - RFC 1939):
  - Ports: Port 110 (plaintext), Port 995 (POP3S encrypted via TLS).
  - Operational Model: Connects, downloads all messages to local computer disk, and deletes them from server mailbox by default.
  - Limitations: Single-device model; read status, sent items, and custom folders do NOT synchronize across other devices (smartphone, office PC).
- Internet Message Access Protocol version 4 (IMAP4 - RFC 3501):
  - Ports: Port 143 (STARTTLS), Port 993 (IMAPS encrypted via TLS).
  - Operational Model: All email messages, read/unread flags, drafts, and folder structures reside permanently on central server.
  - Advantages: Perfect multi-device synchronization; mandatory standard for modern enterprise email.

Email Anti-Spoofing Protocols: SPF, DKIM & DMARC

Combatting Domain Phishing and Fraudulent Impersonation

• The Vulnerability: Standard SMTP headers allow any sender to forge the `From:` address arbitrarily.

• 1. SPF (Sender Policy Framework - RFC 7208):

- Domain owner publishes DNS TXT record listing authorized sending mail server IP addresses.
- Example: `v=spf1 ip4:196.216.180.15 include:_spf.google.com -all`.

• 2. DKIM (DomainKeys Identified Mail - RFC 6376):

- Sending server cryptographically signs message headers with private key; recipient verifies signature using public key in sender's DNS record.

• 3. DMARC (RFC 7489):

- Coordinates SPF and DKIM alignment, instructing receiving servers what to do with failing messages (`none`, `quarantine`, `reject`) and generating forensic telemetry reports.

Academic Typography & Font Engineering

Serif vs Sans-Serif, Kerning, Tracking, and Readability

• Typeface Categories:

- Serif (e.g., Times New Roman, Georgia): Small decorative strokes at letter terminals; guides human eye across horizontal lines; traditional standard for printed body text.
- Sans-Serif (e.g., Arial, Calibri, DejaVu Sans): Clean geometric glyphs without serifs; optimal legibility on digital computer monitors and slide presentations.
- Monospaced (e.g., Courier New, Consolas): Every character occupies identical horizontal width; mandatory for code snippets, hexadecimal addresses, and tabular alignment.

• Micro-Typography:

- Leading (Line Spacing): Vertical distance between baselines (academic standard: 1.5 lines or 12 pt font with 18 pt leading).
- Kerning: Dynamic spacing adjustment between specific character pairs (e.g., 'AV', 'To').

Document Security & The PDF/A Archival Standard

Digital Preservation, Password Protection & Immutability

• Word Document Vulnerabilities: Editable `.docx` files risk accidental formula deletion, font substitution on different operating systems, and malicious macro execution.

• Portable Document Format (PDF) by Adobe (ISO 32000): Preserves exact typographical layout, vector graphics, and embedded fonts across any operating system.

• PDF/A Archival Standard (ISO 19005):

- Specifically designed for 50+ year long-term digital preservation of SUA theses and university charters.
- Strict Rules: Mandates 100% font embedding; forbids external font references, dynamic JavaScript, audio/video streams, and encryption.

• Document Protection Controls: Setting Document Open Passwords (AES-128/256) and Permission Passwords restricting printing or text copying.

Entering and Checking Spreadsheet Data

Simple ways to reduce data-entry mistakes

- Use consistent headings and units for each column.
- Use a drop-down list when the class has a short set of approved labels, such as crop type.
- Check spelling, missing values, and unexpected entries before using a worksheet.
- Keep a protected copy of the original records before editing them.
- Ask a lecturer or data owner when an entry is unclear; do not guess.

Formatting a Readable Worksheet

Presenting records clearly

- Use a clear font and readable column widths.
- Distinguish headings from data using consistent formatting.
- Use borders or light shading sparingly to separate sections.
- Keep colors high-contrast and do not rely on color alone to convey meaning.
- Preview the worksheet before printing or sharing.

Reviewing and Sorting Worksheet Records

Keeping small class datasets understandable

- Sort records by a meaningful field, such as date, animal ID, or crop name.
- Use a filter to temporarily show records that match a label.
- Confirm that each row stays together when sorting.
- Clear filters before reporting the full dataset.
- Save a separate copy before making substantial changes.

Dates and Text in a Worksheet

Entering information consistently without date calculations

- Use one consistent date format throughout a worksheet.
- Label text fields clearly, such as animal ID, crop, or location.
- Avoid mixing dates, notes, and measurements in the same column.
- Check that imported text has not been split or joined incorrectly.
- Use simple headings so another student can understand the records.

Saving and Sharing a Spreadsheet

Protecting class and research records

- Give the file a clear name that identifies the topic and date.
- Save the worksheet in an approved location and make a backup copy.
- Share only with people who are authorized to view the information.
- Avoid putting personal or sensitive information in a public workbook.
- Close or protect the workbook when using a shared computer.

Simple Charts from Worksheet Data

Choosing a clear visual for a small dataset

- Select a small, labeled data range before inserting a chart.
- Use a column chart to compare categories, such as crop types.
- Use a line chart to show a trend across time when observations are ordered.
- Add a title and axis labels so the chart can be understood on its own.
- Check that the selected chart type does not hide important information.

Web Browser Engine Architecture

Parsing HTML, Constructing the DOM, and JavaScript Execution

• Core Subsystems of a Modern Web Browser:

1. User Interface: Address bar, back/forward buttons, bookmarking menu.
2. Browser Engine: Coordinates actions between UI and rendering engine.
3. Rendering Engine: Parses HTML and CSS to render graphical web pages (Blink in Chrome/Edge, Gecko in Firefox, WebKit in Safari).
4. JavaScript Engine: Compiles and executes script code (V8 in Chrome, SpiderMonkey in Firefox).
5. Networking Layer: Handles HTTP/HTTPS, DNS resolution, and TLS handshakes.

• Critical Rendering Path:

- Parses HTML → Builds Document Object Model (DOM) tree.
- Parses CSS → Builds CSS Object Model (CSSOM) tree.
- Combines DOM + CSSOM → Render Tree → Layout phase → Paint phase.

Web State Management: Cookies vs LocalStorage

Maintaining Stateful Sessions Across Stateless HTTP

• HTTP is Stateless: Each HTTP request is completely independent; servers do not remember previous requests without state mechanisms.

• HTTP Cookies (RFC 6265): Small key-value text pairs (max 4 KB) stored by browser and transmitted back to server with EVERY HTTP request header.

- Session Cookies: Stored in volatile RAM; deleted when browser closes.

- Persistent Cookies: Stored on disk until expiration date (e.g., 'Remember Me' logins).

- Security Flags: `Secure` (transmitted only over HTTPS), `HttpOnly` (inaccessible to JavaScript, neutralizing XSS credential theft), `SameSite=Strict` (prevents CSRF attacks).

• HTML5 Web Storage: `localStorage` (persists indefinitely, 5–10 MB, client-side only) and `sessionStorage` (cleared on tab close).

Domain Name System (DNS) Hierarchical Resolution

Translating Human Hostnames into Machine IP Addresses

• The Phonebook of the Internet: Resolves `www.sua.ac.tz` to IPv4 `196.216.180.15`.

• The Hierarchical Tree Architecture:

1. Root DNS Servers (`.`): 13 logical root server clusters managed globally via Anycast routing.
2. Top-Level Domain (TLD) Servers: Manage gTLDs (`.com`, `.org`) and ccTLDs (`.tz`, `.ke`).
3. Authoritative Nameservers: Hold official master DNS records for the organization domain (`sua.ac.tz`).

• Essential DNS Record Types:

- `A` Record: Maps hostname to 32-bit IPv4 address.
- `AAAA` Record: Maps hostname to 128-bit IPv6 address.
- `CNAME` (Canonical Name): Alias mapping one hostname to another.
- `MX` (Mail Exchanger): Identifies mail servers responsible for domain email.
- `TXT`: Arbitrary text; houses SPF, DKIM public keys, and domain ownership tokens.

Transport Layer Security (TLS 1.3) Handshake

Establishing Encrypted, Tamper-Proof HTTPS Sessions

• Core Goals: Confidentiality (AES-GCM encryption), Integrity (HMAC authentication), Server Authenticity (X.509 digital certificate).

• TLS 1.3 Rapid Handshake Sequence (1-RTT Round Trip Time):

1. `ClientHello`: Client sends supported cipher suites and ephemeral Diffie-Hellman public key share.
2. `ServerHello`: Server selects cipher suite, sends its own key share, its X.509 certificate, and digital signature.
3. Session Key Derivation: Both parties use the key-exchange process to establish a shared session key.
4. Encrypted Transmission: All subsequent HTTP requests and responses encrypted with high-speed AES-256-GCM.

• Perfect Forward Secrecy (PFS): Even if the server's master private key is stolen years later, past recorded encrypted sessions cannot be decrypted.

Cloud Productivity Ecosystems in Academic Research

Microsoft 365, Google Workspace, and Private Clouds

• Collaborative Cloud Architecture: Real-time co-authoring using Operational Transformation (OT) and Conflict-Free Replicated Data Types (CRDT).

• Enterprise Cloud Suites:

- Microsoft 365: Word Online, Excel Online, OneDrive, SharePoint; deep enterprise Active Directory integration.
- Google Workspace: Google Docs, Sheets, Drive; streamlined browser-native collaboration.
- Self-Hosted Private Clouds (Nextcloud / ownCloud): Deployed on SUA DIIT servers to ensure sensitive national agricultural research data remains sovereign within Tanzanian borders.

• Cloud Storage Mechanics: Delta sync (transmitting only modified byte blocks of large files) and continuous versioning allowing rollback to previous revisions.

Topic 5: Key Takeaways & Synthesis Matrix

Summary of Applications & Networking for Examination Mastery

• Document Engineering: Section breaks isolate independent formatting; heading styles empower dynamic tables of contents; mail merge automates mass communications.

• Spreadsheet Referencing: Relative (`A1`) shifts dynamically; Absolute (`$A$1`) stays locked; Mixed (`$A1`/`A$1`) locks one coordinate.

• Simple Spreadsheet Work: Cell references, basic arithmetic formulas, `SUM`, and `AVERAGE` support introductory class tasks.

• Web Infrastructure: URLs specify protocol, host, port, path, and query; HTTP request-response cycle handles stateless web interactions.

• Email Architecture: SMTP relays messages server-to-server; IMAP4 provides multi-device synchronized retrieval; SPF/DKIM/DMARC eliminate domain spoofing.

Review Quiz & Knowledge Check

Self-Testing Key Software & Protocol Principles

1. Why is a Section Break mandatory if you need Chapter 4 of a thesis in Landscape and Chapter 5 in Portrait?
2. When copying formula `=$C$2 * D5` from cell E5 to E6, what will the resulting formula be? (Ans: `=$C$2 * D6`).
3. Name a suitable chart for comparing crop types and explain how you would label it.
4. Differentiate between POP3 and IMAP4 regarding multi-device message synchronization.
5. What does the `-all` mechanism at the end of an SPF DNS TXT record instruct receiving mail servers to do?

Reading Assignments & Homework Notice

Preparation for Next Lecture & Continuous Assessment

• Required Readings:

- Microsoft Support. Overview of formulas in Excel. https://support.microsoft.com/en-us/excel/get-started/overview-of-formulas-in-excel
- Smith, J. L., Wang, S., & Mortati, J. - Information Systems for Business and Beyond (2019), Chapter 5: Networking and Communication. https://opentextbook.site/informationsystems2019/

• Homework Assignment 5 Notice:

- Available on SUA Moodle / Assignment directory: `BIT_111_Assignment_5.pdf`.
- Total Allocation: 100 Marks (Covers document structuring, simple spreadsheet use, and email protocols).
- Submission Deadline: End of Week 10 by 17:00 EAT.

• Next Lecture Preview: Topic 6 — Data Security and Malware Defense (Physical Protection, Cryptography, Anti-Malware).

Topic 5: Closeup of Excel Spreadsheet template to track printouts (29911005444).jpg

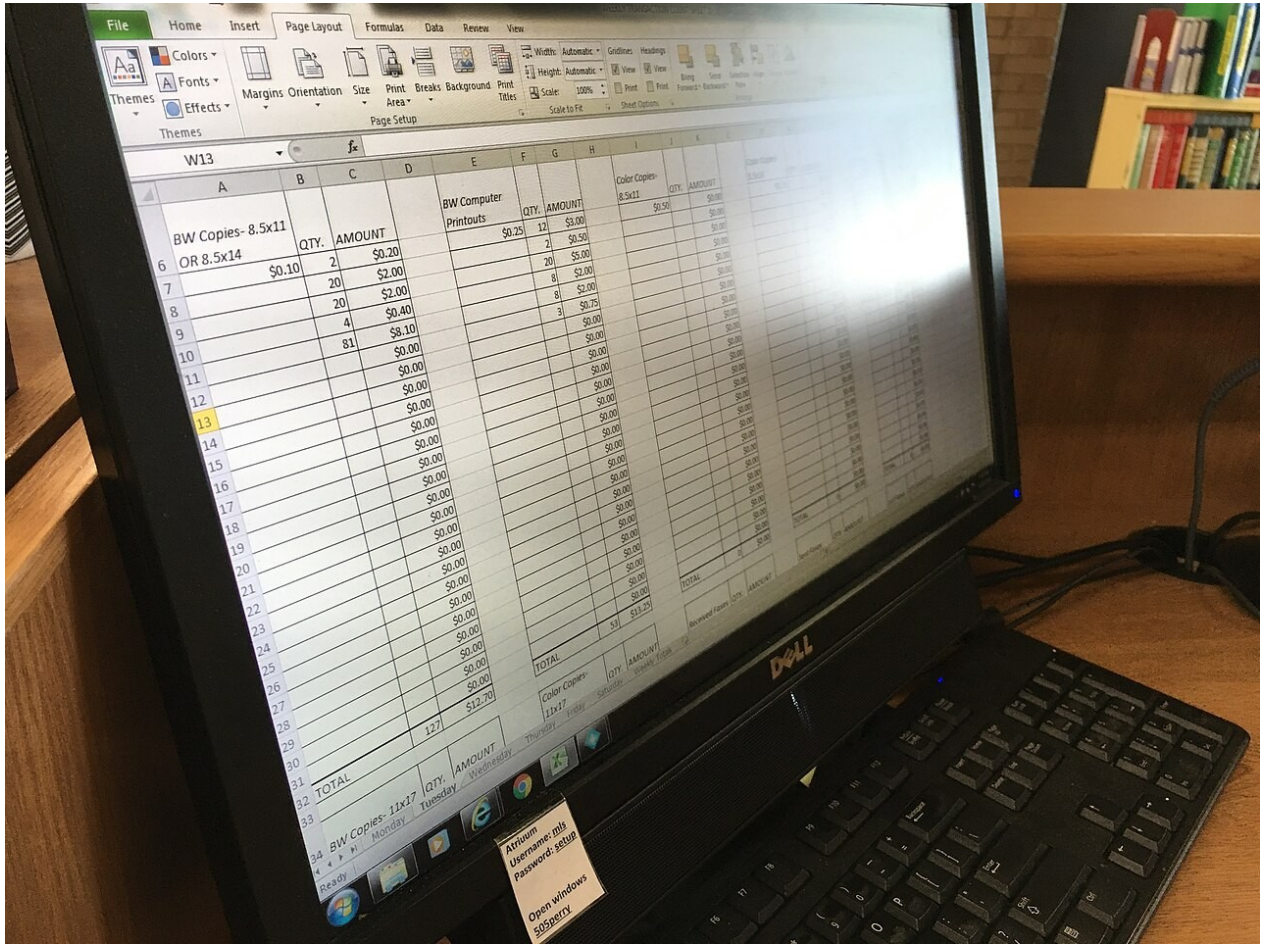


IMAGE NOTES

Description: Closeup of Excel Spreadsheet template to track printouts (29911005444)

Creator: Texas State Library and Archives Commission from Austin, TX, United States

License: CC BY 2.0
Source: Wikimedia Commons file page
License details

Topic 5: Excel chart.PNG

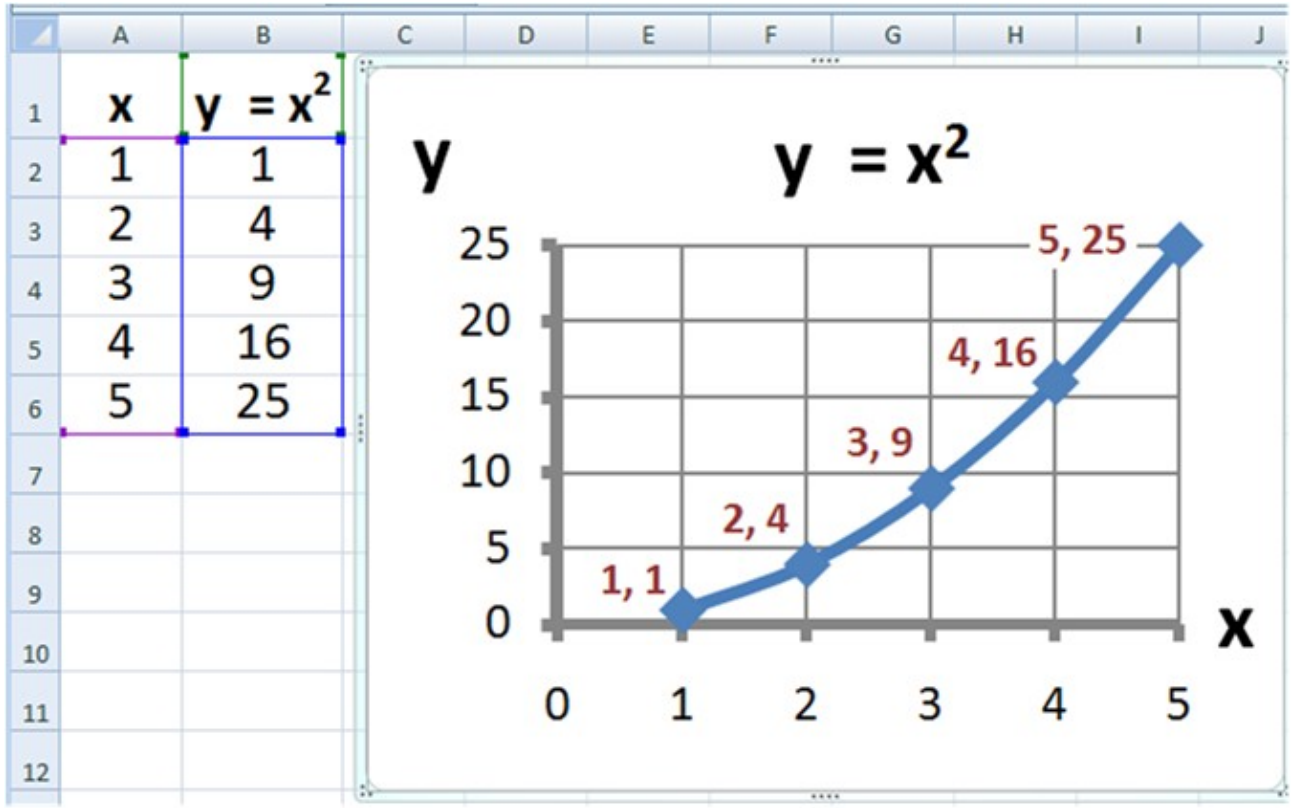


IMAGE NOTES

Description: Excel chart

Creator: Brews ohare

License: Public domain
Source: Wikimedia Commons file page

Topic 5: Fungsi Proper Pada Microsoft Excel.png

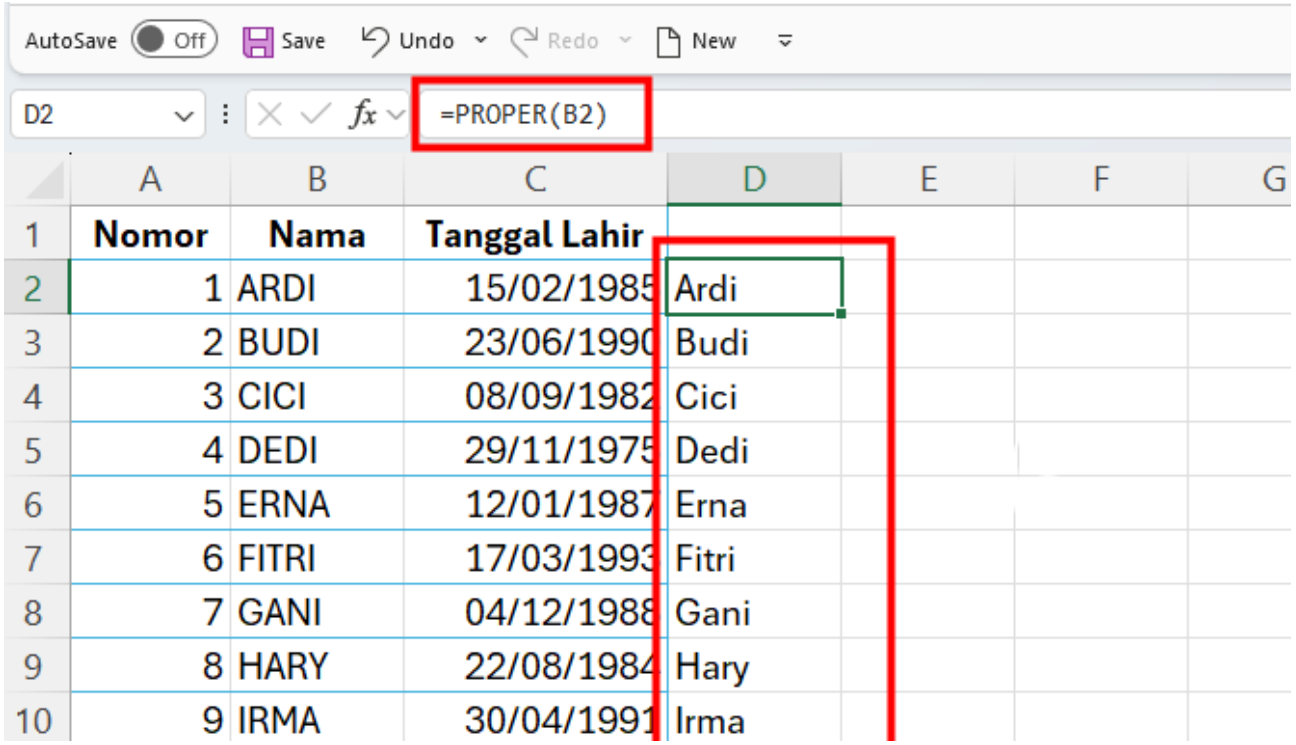


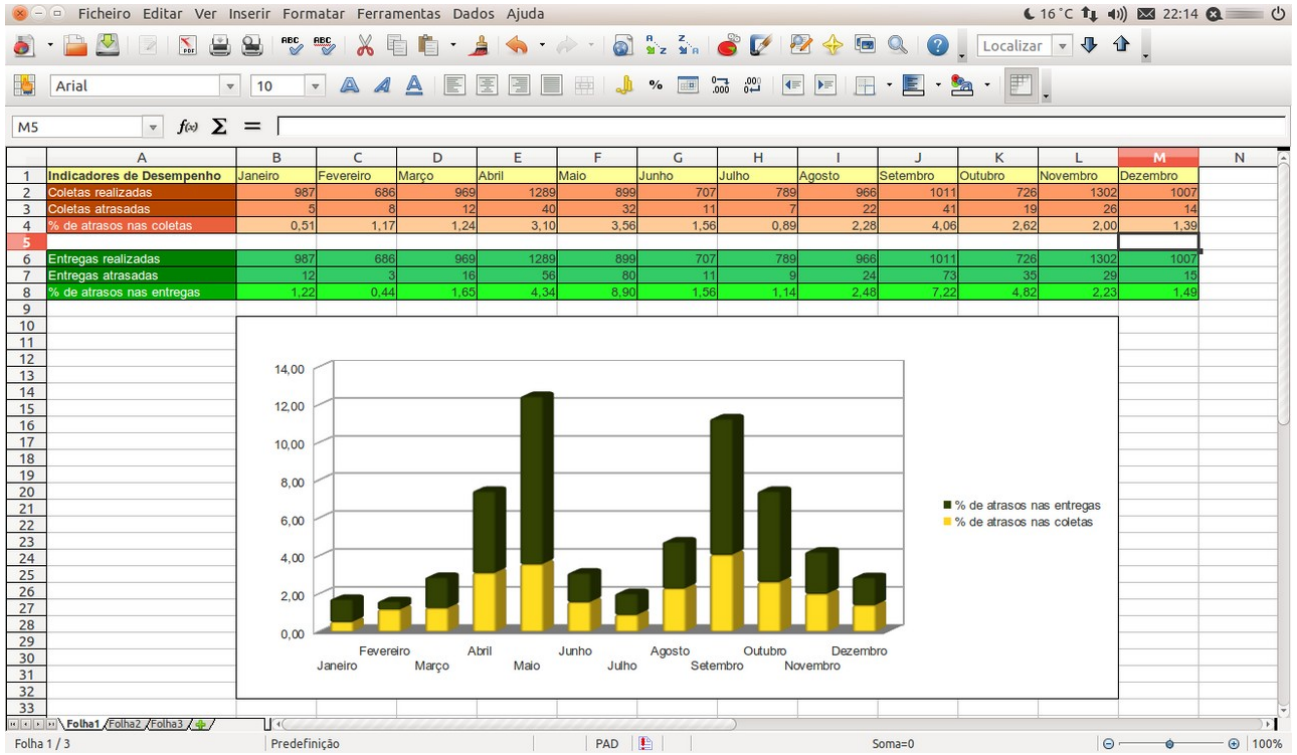
IMAGE NOTES

Description: Fungsi Proper Pada Microsoft Excel

Creator: Umulhidayah

License: CC BY-SA 4.0
Source: Wikimedia Commons file page
License details

Topic 5: Libreoffice-calc.png



	Janeiro	Fevereiro	Março	Abril	Maio	Junho	Julho	Agosto	Setembro	Outubro	Novembro	Dezembro
Coletas realizadas	987	686	969	1289	899	707	789	966	1011	726	1302	1007
Coletas atrasadas	5	8	12	40	32	11	7	22	41	19	26	14
% de atrasos nas coletas	0,51	1,17	1,24	3,10	3,56	1,56	0,89	2,28	4,06	2,62	2,00	1,39
Entregas realizadas	987	686	969	1289	899	707	789	966	1011	726	1302	1007
Entregas atrasadas	12	3	16	56	80	11	9	24	73	35	29	15
% de atrasos nas entregas	1,22	0,44	1,65	4,34	8,90	1,56	1,14	2,48	7,22	4,82	2,23	1,49

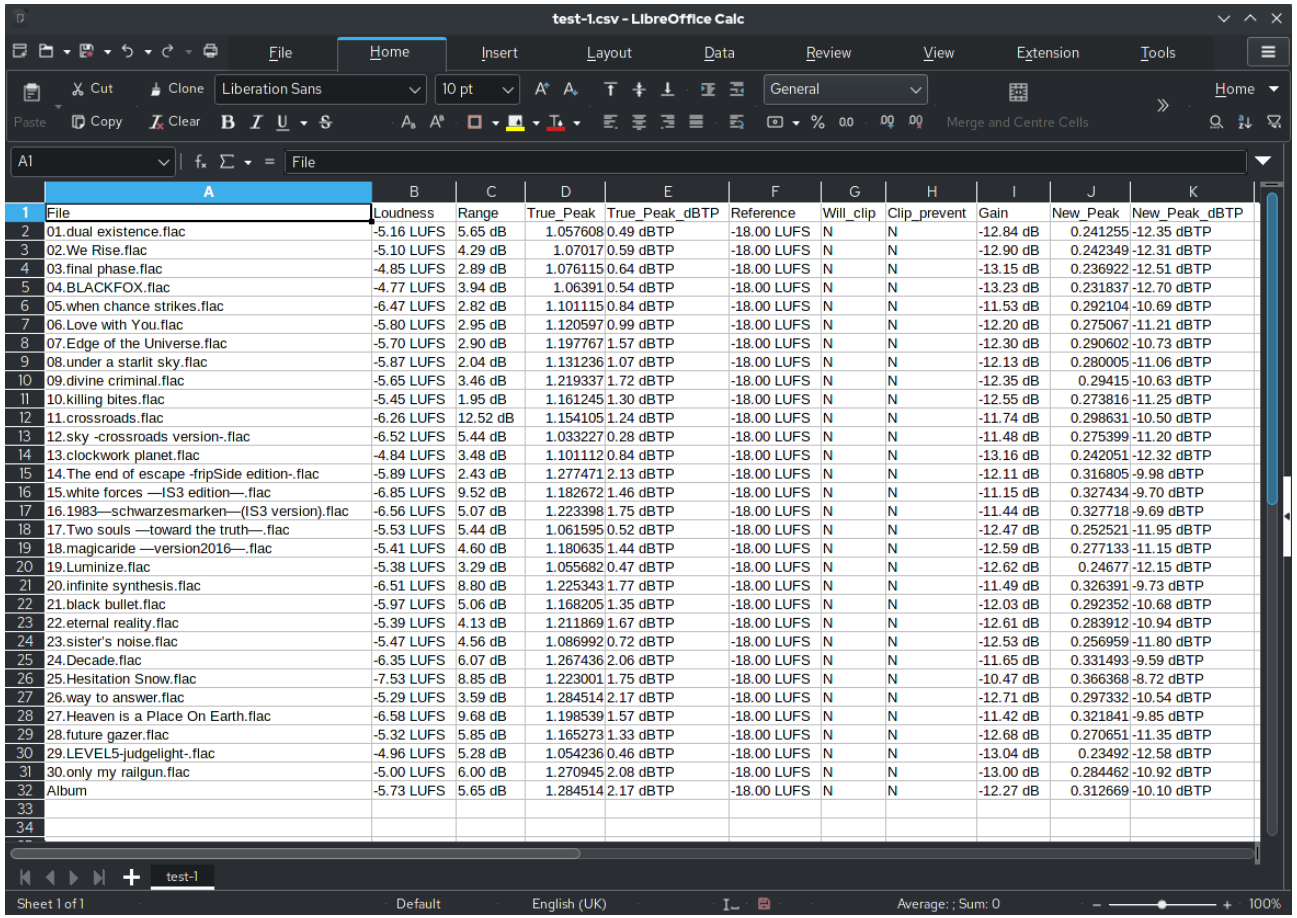
IMAGE NOTES

Description: Libreoffice-calc

Creator: The Document Foundation and Canonical

License: CC0
Source: Wikimedia Commons file page
License details

Topic 5: LibreOffice 7.2.4.1 Calc with csv screenshot.png



	A	B	C	D	E	F	G	H	I	J	K
1	File	Loudness	Range	True_Peak	True_Peak_dBTP	Reference	Will_clip	Clip_prevent	Gain	New_Peak	New_Peak_dBTP
2	01.dual existence.flac	-5.16 LUFS	5.65 dB	1.057608	0.49 dBTP	-18.00 LUFS	N	N	-12.84 dB	0.241255	-12.35 dBTP
3	02.We Rise.flac	-5.10 LUFS	4.29 dB	1.07017	0.59 dBTP	-18.00 LUFS	N	N	-12.90 dB	0.242349	-12.31 dBTP
4	03.final phase.flac	-4.85 LUFS	2.89 dB	1.076115	0.64 dBTP	-18.00 LUFS	N	N	-13.15 dB	0.236922	-12.51 dBTP
5	04.BLACKFOX.flac	-4.77 LUFS	3.94 dB	1.06391	0.54 dBTP	-18.00 LUFS	N	N	-13.23 dB	0.231837	-12.70 dBTP
6	05.when chance strikes.flac	-6.47 LUFS	2.82 dB	1.101115	0.84 dBTP	-18.00 LUFS	N	N	-11.53 dB	0.292104	-10.69 dBTP
7	06.Love with You.flac	-5.80 LUFS	2.95 dB	1.120597	0.99 dBTP	-18.00 LUFS	N	N	-12.20 dB	0.275067	-11.21 dBTP
8	07.Edge of the Universe.flac	-5.70 LUFS	2.90 dB	1.197767	1.57 dBTP	-18.00 LUFS	N	N	-12.30 dB	0.290602	-10.73 dBTP
9	08.under a starlit sky.flac	-5.87 LUFS	2.04 dB	1.131236	1.07 dBTP	-18.00 LUFS	N	N	-12.13 dB	0.280005	-11.06 dBTP
10	09.divine criminal.flac	-5.65 LUFS	3.46 dB	1.219337	1.72 dBTP	-18.00 LUFS	N	N	-12.35 dB	0.29415	-10.63 dBTP
11	10.killing bites.flac	-5.45 LUFS	1.95 dB	1.161245	1.30 dBTP	-18.00 LUFS	N	N	-12.55 dB	0.273816	-11.25 dBTP
12	11.crossroads.flac	-6.26 LUFS	12.52 dB	1.154105	1.24 dBTP	-18.00 LUFS	N	N	-11.74 dB	0.298631	-10.50 dBTP
13	12.sky -crossroads version-.flac	-6.52 LUFS	5.44 dB	1.033227	0.28 dBTP	-18.00 LUFS	N	N	-11.48 dB	0.275399	-11.20 dBTP
14	13.clockwork planet.flac	-4.84 LUFS	3.48 dB	1.101112	0.84 dBTP	-18.00 LUFS	N	N	-13.16 dB	0.242051	-12.32 dBTP
15	14.The end of escape -fripSide edition-.flac	-5.89 LUFS	2.43 dB	1.277471	2.13 dBTP	-18.00 LUFS	N	N	-12.11 dB	0.316805	-9.98 dBTP
16	15.white forces —IS3 edition—.flac	-6.85 LUFS	9.52 dB	1.182672	1.46 dBTP	-18.00 LUFS	N	N	-11.15 dB	0.327434	-9.70 dBTP
17	16.1983—schwarzesmarken—(IS3 version).flac	-6.56 LUFS	5.07 dB	1.223398	1.75 dBTP	-18.00 LUFS	N	N	-11.44 dB	0.327718	-9.69 dBTP
18	17.Two souls —toward the truth—.flac	-5.53 LUFS	5.44 dB	1.061595	0.52 dBTP	-18.00 LUFS	N	N	-12.47 dB	0.252521	-11.95 dBTP
19	18.magicaride —version2016—.flac	-5.41 LUFS	4.60 dB	1.180635	1.44 dBTP	-18.00 LUFS	N	N	-12.59 dB	0.277133	-11.15 dBTP
20	19.Luminize.flac	-5.38 LUFS	3.29 dB	1.055682	0.47 dBTP	-18.00 LUFS	N	N	-12.62 dB	0.24677	-12.15 dBTP
21	20.infinite synthesis.flac	-6.51 LUFS	8.80 dB	1.225343	1.77 dBTP	-18.00 LUFS	N	N	-11.49 dB	0.326391	-9.73 dBTP
22	21.black bullet.flac	-5.97 LUFS	5.06 dB	1.168205	1.35 dBTP	-18.00 LUFS	N	N	-12.03 dB	0.292352	-10.68 dBTP
23	22.eternal reality.flac	-5.39 LUFS	4.13 dB	1.211869	1.67 dBTP	-18.00 LUFS	N	N	-12.61 dB	0.283912	-10.94 dBTP
24	23.sister's noise.flac	-5.47 LUFS	4.56 dB	1.086992	0.72 dBTP	-18.00 LUFS	N	N	-12.53 dB	0.256959	-11.80 dBTP
25	24.Decade.flac	-6.35 LUFS	6.07 dB	1.267436	2.06 dBTP	-18.00 LUFS	N	N	-11.65 dB	0.331493	-9.59 dBTP
26	25.Hesitation Snow.flac	-7.53 LUFS	8.85 dB	1.223001	1.75 dBTP	-18.00 LUFS	N	N	-10.47 dB	0.366368	-8.72 dBTP
27	26.way to answer.flac	-5.29 LUFS	3.59 dB	1.284514	2.17 dBTP	-18.00 LUFS	N	N	-12.71 dB	0.297332	-10.54 dBTP
28	27.Heaven is a Place On Earth.flac	-6.58 LUFS	9.68 dB	1.198539	1.57 dBTP	-18.00 LUFS	N	N	-11.42 dB	0.321841	-9.85 dBTP
29	28.future gazer.flac	-5.32 LUFS	5.85 dB	1.165273	1.33 dBTP	-18.00 LUFS	N	N	-12.68 dB	0.270651	-11.35 dBTP
30	29.LEVEL5-judgelight-.flac	-4.96 LUFS	5.28 dB	1.054236	0.46 dBTP	-18.00 LUFS	N	N	-13.04 dB	0.23492	-12.58 dBTP
31	30.only my railgun.flac	-5.00 LUFS	6.00 dB	1.270945	2.08 dBTP	-18.00 LUFS	N	N	-13.00 dB	0.284462	-10.92 dBTP
32	Album	-5.73 LUFS	5.65 dB	1.284514	2.17 dBTP	-18.00 LUFS	N	N	-12.27 dB	0.312669	-10.10 dBTP
33											
34											

IMAGE NOTES

Description: LibreOffice 7.2.4.1 Calc with csv screenshot

Creator: The Document Foundation, Alessandro Ghedini, Matthias C. Hormann, and contributors; screenshot: VulcanSphere

License: MPL 2
Source: Wikimedia Commons file page
License details

Topic 5: SUM formula in LibreOffice Calc.png

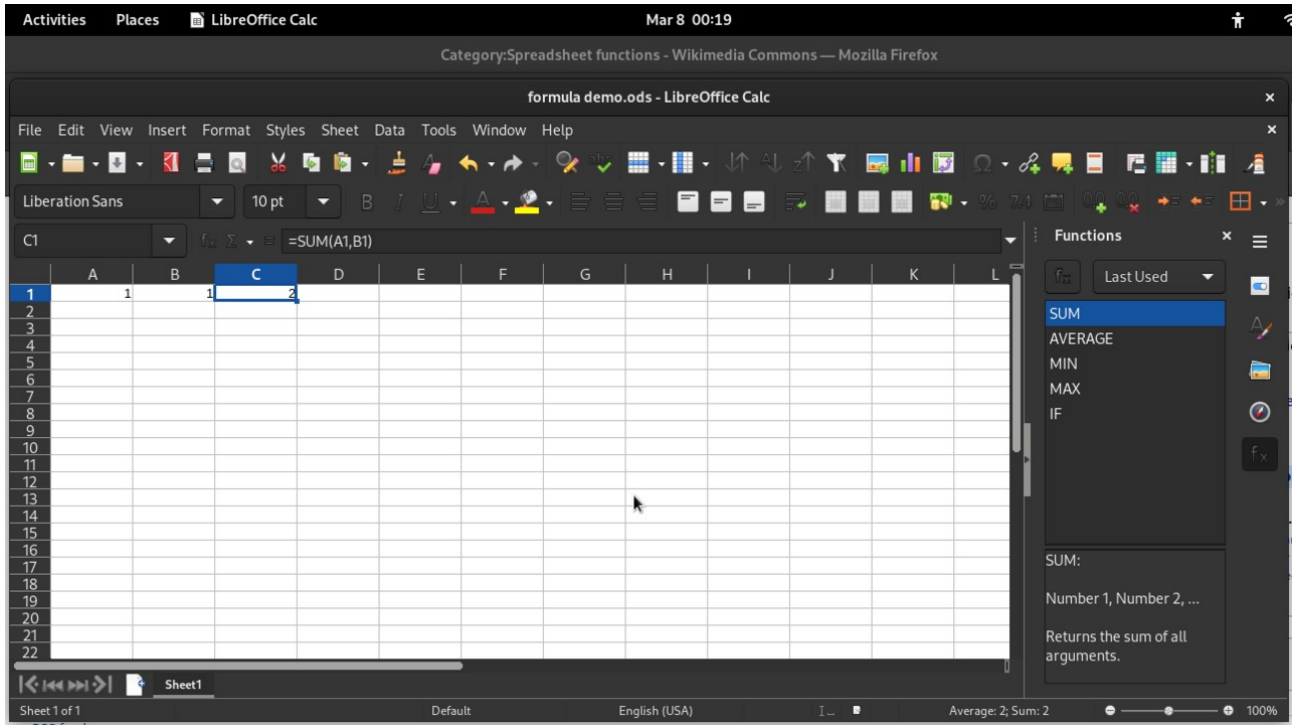


IMAGE NOTES

Description: SUM formula in LibreOffice Calc

Creator: Arlo Barnes

License: CC0
Source: Wikimedia Commons file page
License details

Topic 5: Mozilla Thunderbird 140 ESR on Debian 13 (Trixie) and KDE Plasma 6.png

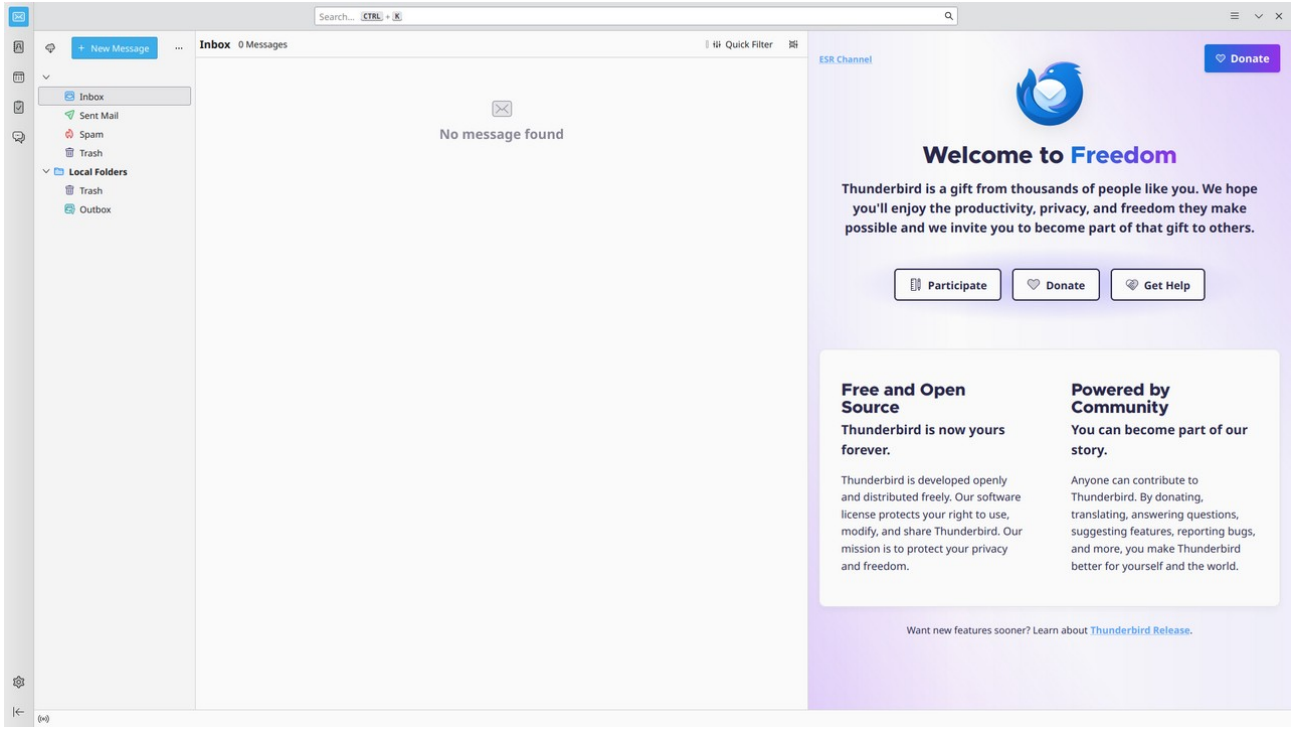


IMAGE NOTES

Description: Mozilla Thunderbird 140 ESR on Debian 13 (Trixie) and KDE Plasma 6

Creator: Victor Matheus Amaral

License: CC0
Source: Wikimedia Commons file page
License details

Topic 5: CPT 8100 Word Processor Desktop Microcomputer 5185A65A.png



IMAGE NOTES

Description: CPT 8100 Word Processor Desktop Microcomputer 5185A65A

Creator: LehmanUMN

License: CC0
Source: Wikimedia Commons file page
License details